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Urinary incontinence – classification

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Definition

In developed countries, people aged 64 years old and older are the fastest growing population. Urinary incontinence is a wide-spread disease and rises with aging. Complaints range from sporadic, drop-wise urine loss to more severe urine loss with a feeling of insecurity (smell, wetness), to social withdrawal. The personal and socio-economic burden is highly underestimated.

Urinary incontinence is defined as the complaint of any involuntary leakage of urine (International Continence Society, ICS 2002). Incontinence is not regarded as one entity. It has to be distinguished according to symptoms and underlying diseases, such as infections, etc., they have to be excluded (see below).

Most women do not speak to their doctor about their incontinence. This may be influenced by the widespread opinion that incontinence is a normal condition after childbirth or occurs with aging. However, a radical change is underway. Understandably, the demands of women affected by this condition have to be addressed. Nowadays, women want to remain their professional and private activities and their mobility. They wish to keep their continence up to elderly age. Even retired women do not want to give up their walks and "physically challenging" hobbies.

Definitions by ICS & IUGA

The International Continence Society (ICS) and the International Urogynecological Association (IUGA) set out the definitions of incontinence in joint reports. They take into account the clinical character and symptoms, as given by the patient.

The ICS and IUGA are constantly attempting to standardize the key terms, units, and examination tools to be used for describing continence and pelvic floor issues, and makes all documents relating to standardization freely available on the website icsoffice.org.

International Continence Society (ICS)

International Urogynecological Association (IUGA)

Most common forms of incontinence in women

Urinary incontinence is defined as complaints of involuntary loss of urine.

In general, there are two main forms of incontinence in women [B05]:

- Stress urinary incontinence: Involuntary loss of urine during exertion or physical activity (e.g. during sport) or when sneezing or coughing.
- Overactive bladder (also called OAB (wet) or urge incontinence): Involuntary loss of urine on urge. Overactive bladder can also occur without incontinence (OAB dry). The key difference is, whether or not the toilet can be reached in time when an urge to urinate occurs - this is also dependent on the situation (at home, away from home, etc.).
  [B05]

Mixed urinary incontinence, which is more common in older patients, refers to the combination of both symptoms of stress incontinence and overactive bladder. [P03]

Urine incontinence

Incontinence is a storage function pathology. The various forms of incontinence have either a functional or an anatomical cause. The above mentioned main forms of incontinence are by far the most frequently encountered in GP and gynecology incontinence consultations (90 %).

Epidemiologically, stress urinary incontinence is the most common, with a prevalence of up to approx. 80 % for geriatric people. Overactive bladder with urge incontinence (OAB wet) makes up between 10 and 30 %, while the proportion of mixed urinary incontinence, i.e. a combination of stress urinary incontinence and overactive bladder, is between 20 and 60 %.

Stress urinary incontinence (SUI)

During physical effort, such as sneezing or coughing, involuntary urine leakage occurs without an urge to urinate. In general, any mechanical stress that leads to an increase in abdominal pressure can trigger stress urinary incontinence. In this case, the urethral internal sphincter opens in an uncontrolled manner, without any simultaneous detrusor contraction.

As the pressure in the bladder only increases slightly as it fills, due to the compliance of the bladder walls, in phases of physical relaxation only a slightly positive pressure gradient in the urethra is required to prevent involuntary urine leakage. This situation changes under physical stress (e.g. sneezing and coughing), which can lead to a rapid, significant increase in intra-abdominal pressure and easily exceeds even the best urethral closure pressure.

Two mechanisms are required to maintain continence:

- "Guarding reflex" which actively increases the muscle tone of the urethral sphincter muscles in these situations
- Passive transmission of the increase in intra-abdominal pressure to the urethra through the slow twitch fibres

The more this pressure transmission declines, the greater the probability of stress urinary incontinence arises. Conversely, excellent pressure transmission is required if the individual urethral closure pressure is very low. Pressure transmission is dependent on the tone of the pelvic floor muscles and the elasticity of the connective tissue. The urethral sphincter muscles are striated skeletal muscles and can be trained accordingly with physiotherapy. [P05]

Main causes of stress urinary incontinence: damage to the bladder and urethra supporting structures due to pregnancy and childbirth, or the aging process.

In the past, significant damage of the supporting structures and the resulting urethral (hyper) mobility as a cause of stress urinary incontinence was overestimated. However, hormonal and neuronal factors also play a role in continence.

In addition to these factors, there are other risk factors such as parity, body weight and smoking.

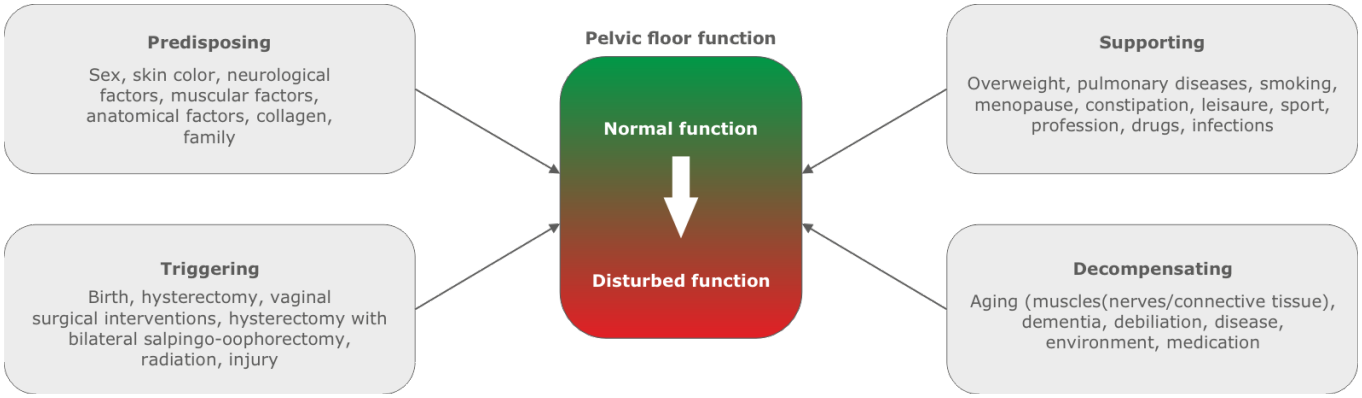

Figure 1: Risk factors for pelvic floor dysfunction.

Overactive bladder (OAB)

Synonyms: hyperactive bladder, hyperactive urinary bladder syndrome, (the term “irritable bladder syndrome” is also sometimes used in the general press).

In 2010, the ICS/IUGA rolled out a new definition for the condition previously known as urge incontinence or irritable bladder.

Since then, “Overactive Bladder (OAB)”, or overactive bladder syndrome, has been the preferred term. This is a symptom-oriented paraphrasing of a condition. Urgency is defined as a key symptom. It is understood to be an intense, sudden, urgent need to urinate. This urge can hardly be suppressed with effort. However, “urge” can also be understood as a “normal” urge to urinate. An urge to urinate without involuntary urine leakage is called OAB dry. In this case, the urethral sphincter is still strong enough to withstand the detrusor contraction with sufficient pressure. In the case of "urge incontinence", this is no longer guaranteed, i.e. involuntary loss of urine occurs when an imperative urge occurs (OAB wet - "wet" hyperactive bladder).
Frequency (urination > 8 times in 24 hours), reduced urination volume and nocturia are a consequence of the imperative urge to urinate. In one third of patients with OAB, also suffer from urge incontinence.

For the diagnosis of overactive bladder, the ICS requires to rule out metabolic and local pathological factors (e.g. diabetes mellitus, urinary tract infection, genital atrophy) that could explain these symptoms. However, the ruling out “of other local pathological and endocrinal factors” is not defined precisely. Urine testing and potentially blood testing (glucose) are certainly included. From an urogynecological perspective, the assessment of the pelvic floor including genital atrophy (pH strip test), pelvic organ prolapse and elevated residual volume.

The cause of overactive bladder syndrome is not yet fully understood. In most women an underlying cause cannot be found. This condition is defined as "idiopathic" overactive bladder.

During urodynamics examination, two different kinds of OAB can be distinguished: with or without uninhibited detrusor contractions. Women with bladder overactivity without incontinence (OAB/dry) can show detrusor contractions on urodynamic studies too. In this case, it can be assumed that the urethral sphincter is sufficiently strong to counteract the loss of urine through the fast twitch fibres.

Mixed urinary incontinence

The definition of mixed urinary incontinence from the International Continence Society (ICS) 2002 bases on history and clinical findings alone. The diagnosis can be drawn if women present with symptoms of both overactive bladder and stress urinary incontinence (mixed urinary incontinence = “the complaint of involuntary leakage associated with urgency and also with exertion, effort, sneezing or coughing”) [B01].

Women with mixed urinary incontinence have a higher burden of psychological issues compared to women who suffer from OAB or SUI alone. [B01]

With urodynamic studies the pre-dominant incontinence component can be detected. Urodynamically, women with urge-dominant mixed incontinence present more frequently with detrusor overactivity, early first urge to urinate (< 150 ml bladder volume), and lower maximum bladder capacity (< 300 ml bladder volume). This is reflected in the bladder diary in lower micturition volumes, lower than those of women with pre-dominant stress urinary incontinence. Urodynamic studies are therefore helpful to distinguish between the two conditions, especially the OAB component, while stress urinary incontinence can usually be detected and classified from the medical history and clinical diagnostics, such as a cough test. Urodynamic studies of most women with mixed urinary incontinence do not show stress urinary incontinence and detrusor overactivity simultaneously. In a group of 950 incontinent women at Duke University, 52 % had mixed symptoms, but only 13 % showed mixed findings [P01].

According to Minassian, the prevalence of mixed urinary incontinence is approximately 17 times higher than expected (stress urinary incontinence 13 %, urge incontinence 5 %: Expected: 0,13 x 0,05 = 0,0065, but “Mixed UI found” = 11 %) [P06]. One explanation might be that the called “susceptibility model”, where the presence of one disease supports a second disease. However, the severity of urinary incontinence could also promote symptoms of urgency (“severity model”). A connection between increasing severity of stress urinary incontinence and increasing symptoms of urgency was already suggested by Bump (2003). He hypothesized that mixed urinary incontinence is actually a “severe form” of SUI and/or urge incontinence [P01].

Telemann (2004) was able to show that the frequency of symptoms of urgency increases with the severity of stress incontinence [P02]. [P01][P02][P06]

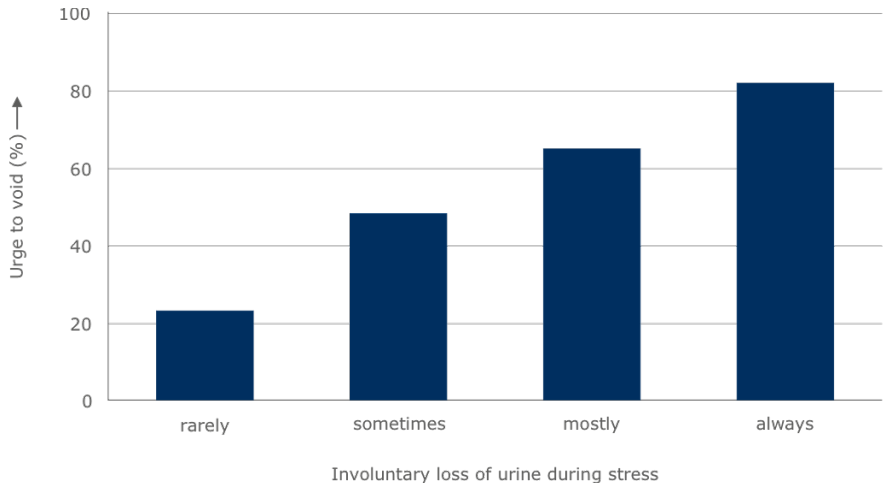

Figure 2: Stress urinary incontinence. Severity and frequency of urgency.
Urodynamics should be indicated if treatment has failed after basic OAB diagnostics (basic diagnostics = medical history, exclusion of urinary tract infection and exclusion of residual urine > 100 ml).

Rare forms of incontinence in women

Incontinence with neurogenic overactive bladder/neurogenic urge incontinence

Women with underlying neurological conditions such as Parkinson's disease, multiple sclerosis and diabetes mellitus, as well as cerebral sclerosis and apoplexy, bladder voiding disorders with urinary incontinence can be observed quite often. Involuntary contractions occur similar to women with idiopathic overactive bladder.

This kind of urine leakage is referred to as incontinence with neurogenic detrusor overactivity, which should replace the older term "reflex incontinence".

In case of urodynamically verified neurogenic detrusor overactivity, there is always a underlying neurological problem.

Chronic retention of urine (previously overflow bladder)

Most often a consequence of long-term distension of the bladder, either due to an intravesicular obstruction (e.g. prolapse, obstructive incontinence surgery) or as a consequence of a peripheral bladder denervation (e.g. radical pelvic surgery). Here, the overfilled bladder presents an intra-vesicular pressure which exceeds the urethral closure pressure even when the body is relaxed. This results in a continuous ("overflow") urine leakage.

Fistula formation

This can occur as bladder or urethral fistula and is either the result of surgical or obstetric injury, inflammatory or radiogenic processes or a tumor-related disease. In rare cases, malformations in the genital tract region cause incontinence.

"Pseudoincontinence"

In terms of differential diagnosis, there are situations that resemble organic incontinence:

- limited mobility → treatment: treat the underlying disease: e.g. arthritis, Parkinson's.
- with poor vision → treatment: simplify access to toilet.

Giggle incontinence

Giggle incontinence is triggered by bursts of laughter and is most probably associated with situational detrusor overactivity, which is almost impossible to reproduce for urodynamics. There is a delayed maturation process at play, as this form of incontinence occurs very rarely in women older than 25 years. A relative estrogen deficiency, for example during puberty and adolescence, might be one explanation.

Enuresis

Enuresis is intermittent incontinence during sleep without an organic cause in people > 5 years of age.

Literature

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B01]	The standardisation of terminology of lower urinary tract function: report from the Standardisation Sub-committee of the International Continence Society	Abrams P, Cardozo L, Fall M, Griffiths D, Rosier P, Ulmstein U, van Kerrebroeck P, Victor A, Wein A	Neurourol Urodyn	21	167-178	200
[B02]	Structural support of the urethra as it relates to stress urinary incontinence: the hammock hypothesis	DeLancey JO	Am J of Obstet Gynecol	170	1713	199
[B03]	Urinary bladder contraction and relaxation: physiology and pathophysiology	Andersson KE, Arner A	Physiol Rev	84	935-986	200

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B04]	Vaginal Birth and de Novo Stress Incontinence: Relative Contributions of Urethral Dysfunction and Mobility	DeLancey JO, Miller J, Kearney R, Howard D, Reddy P, Umek W, Guire KE, Margulies RU, Ashton-Miller JA	Obstet Gynecol	110	354-362	200
[B05]	An International Urogynecological Association (IUGA)/ International Continence Society (ICS) joint report on the terminology for female pelvic floor dysfunction	Haylen B et al.	International Urogynecology Journal			201

Nr.	Title	Author	Journal	Volume	Pages	Yea
[P01]	Duloxetine Urinary Incontinence Study Group. Mixed urinary incontinence symptoms: urodynamic findings, incontinence severity, and treatment response	Bump RC, Norton PA, Zinner NR, Yalcin I	Obstet Gynecol	Jul;102(1)	76-83	200

Nr.	Title	Author	Journal	Volume	Pages	Yea
[P02]	Overactive bladder: prevalence, risk factors and relation to stress incontinence in middle-aged women	Teleman PM, Lidfeldt J, Nerbrand C, Samsioe G, Mattiasson A	BJOG	Jun;111(6)	600-604	200
[P03]	Mixed incontinence: definitions, outcomes, and interventions	Dmochowski R, Staskin D	Curr Opin Urol	15	374–379	200
[P04]	Norwegian EPINCONT study. Epidemiology of Incontinence in the County of Nord-Trondelag. A community-based epidemiological survey of female urinary incontinence: the Norwegian EPINCONT study. Epidemiology of Incontinence in the County of Nord-Trondelag	Hannestad YS, Rortveit G, Sandvik H, Hunskaar S	J Clin Epidemiol	53	1150–1157	200

Nr.	Title	Author	Journal	Volume	Pages	Yea
[P05]	Stress Urinary Incontinence: Relative Importance of Urethral Support and Urethral Closure Pressure	DeLancey JO, Trowbridge ER, MDa, Miller JM, Morgan DM, MDa, Guire K, Fenner D, Weadock WJ, Ashton-Miller JA	J Urol	June 179(6)	2286–2290	200
[P06]	Why do stress and urge incontinence co-occur much more often than expected?	Minassian VA, Stewart WF, Hirsch AG	Int Urogynecol J Pelvic Floor Dysfunct	Oct;19(10)	1429-1440	200